

Storage Types

- Examples of data stored in the cloud include the following:
 - **Office and media files** : Documents, images, videos, and presentations stored online. //
 - **Applications** : Software and services that run using cloud storage. //
 - **Websites** : Website files such as HTML, CSS, images, and scripts. //
 - **Logs** : Records that store system and application activities. //
 - **Databases** : Structured data stored and managed in the cloud. //
 - It is very important to keep data safe in the cloud while ensuring that it is accessible by those authorized to use.
- Data must be protected with security methods, but allowed users should still be able to access it easily.

On-Prem Storage Technologies (1 of 6)

- On-Prem Storage Types
 - **DAS (direct attached storage)**
Storage that is directly connected to the computer or server that uses it.
 - **NAS (network attached storage)**
Uses the network to provide file access to users and systems.
 - **SAN (storage area network)**
A special part of an organization's network used only for storage hardware, also called a storage array.
 - SANs are more complex and expensive to set up and maintain than other storage options. They need more planning, management, and higher costs.
 - A SAN requires specialized switches that connect to servers using **HBAs (host bus adapters)** instead of NICs.
- These switches are designed only for storage traffic.
- More recent SAN technologies use **CNAs (converged network adapters)** or **UTAs (unified target adapters)**
- These adapters combine different network and storage functions.

On-Prem Storage Technologies (3 of 6)

- Redundant Array of Independent Disks (RAID)
 - **RAID (Redundant Array of Independent Disks)**
A group of multiple disks used together as one system to improve **data safety, performance**, or both by using redundancy.
 - The following are the different types of RAID:
 - **RAID 0**
Data is split into pieces and stored on different drives (no redundancy).
 - **RAID 1**
Data is stored on two drives in duplicate for redundancy.
 - **RAID 5**
Data is split into pieces, added parity inf. and stored across three or more drives.
 - **RAID 6**
Similar to RAID 5, data is striped across multiple disks with extra parity for better data recovery.
 - **RAID 10**
Combines **RAID 0** and **RAID 1** by mirroring data first and then striping it for speed and safety.

On-Prem Storage Technologies (6 of 6)

- On-Prem Storage Performance

– The following technologies improve on-prem storage performance:

- **High-speed connections – NVMe-oF (nonvolatile memory express over fabrics)**

Provides very fast read and write performance, similar to SSDs in a SAN.

- **Virtualization – SDS (software-defined storage)**

Virtualizes the storage infrastructure to provide scalable and manageable storage on generic hardware.

- **Hyperconvergence – HCI (hyperconverged infrastructure)**

Combines software-defined compute, networking, and storage services into one system.

Cloud Storage Technologies (2 of 3)

- **File System Storage**

– Works well for organizing files in a small department and for simple file sharing

– Does not scale well when there is a large amount of different data

– Advantage is that metadata can be stored with each file.

- **Block Storage**

– Each block is addressed by a unique number called a **LUN**, not by file names

– **Block zoning** allows different disk zones to store different data types and helps reduce latency.

Cloud Storage Technologies (3 of 3)

- **Object Storage**

– A cloud-native storage type that uses the cloud's auto-scaling to store large amounts of data

– Uses a flat namespace where all objects are stored in one storage pool

- **Choosing Between File, Block, and Object Storage**

– Object storage maximizes storage capabilities in the cloud

– Object storage is best for data that is written once and then mostly read

– Block storage works well for databases, operating systems, and VM-attached storage

– File system storage is used for organized and managed file systems

Storage Migrations

- The following list explores factors to consider when migrating data from on-prem to the cloud:

- **Migration plan**

Decide which data should be moved first.

- **Data transfer costs**

Moving data into the cloud is usually free, unless a special migration service is used.

- **Data transfer time**

Moving large amounts of data can take time and cause synchronization problems.

- **Data types**

Different data types may need different storage services:

- Files stored in hierarchical file systems
- Volumes managed by block storage
- Unorganized data stored in object storage services

Storage Optimization Techniques

- Storage performance can be evaluated using measurable factors:

- **Transfer rate or read/write throughput**

Shows how fast data can be read from or written to storage. //

- **IOPS (input/output operations per second)**

Measures how many read and write operations storage can perform per second. //

- **Latency**

The time it takes for a storage system to respond to a request. //

Storage Capacity

- The **TP (thin provisioning)** model is an approach where

Users are allocated more storage space than what is physically available in the storage system.

- The **FP (fat provisioning)** model, (also called **thick provisioning**) is an approach where

Only existing physical storage space can be allocated to users.

- When attaching a storage volume to a VM Most public cloud platforms require full allocation of storage space even if it is not fully used.

Data Optimization

- **Compression**

Reduces the number of bits needed to store data by encoding it more efficiently.

- The following are examples of compression technologies:

- Zip compressors such as 7-Zip, xz, and gzip

- RAR compressors such as WinRAR or 7-Zip

- Another method to reduce stored data size is **data deduplication**, which removes multiple copies of the same data.

Data Lifecycle (1 of 4)

- **Tiering**

- **Storage tiers**, also called storage classes, reduce storage cost by increasing the cost and time to retrieve data
- Frequently accessed data can be stored in a service that charges more for storage space and less for access

- **Lifecycle Rules**

- In AWS S3, **lifecycle rules** can be set for one or more objects in a bucket
- These rules can move objects or object versions between storage classes after a certain number of days

Data Lifecycle (3 of 4)

- **Versioning**

- **Versioning** means keeping older versions of data as backup or to track changes for compliance
- A **record** is data that provides evidence of business activities and decisions
- When data is marked as a record, retention-based restrictions are applied

- **Legal Hold**

- **Records management** is the process of identifying and protecting data that qualifies as records
- A **legal hold** is a court-ordered rule that prevents deletion of specific data

Data Lifecycle (4 of 4)

- **Data Destruction**

- Destroying cloud data can be difficult because copies of the data may exist on personal devices in a **hybrid cloud storage** model
- The following options help increase the security of cloud-based data disposal:
 - **Use restrictions**
 - **Tenancy** – Limits data storage to a single host or a single customer
 - **Encryption** (if the key deleted, data can't be used)

Edge Computing

- **Edge computing**

Is an evolution of cloud computing that moves compute power, storage, and connectivity **closer to where data is used**. ---> **low latency , faster response , low workload for main server.**

- **A CDN (content delivery network)**

Is a distributed storage system that stores files closer to users' locations.

- **The CDN service**

Sends user traffic to the closest edge location with the best performance at request time.

- **Parameters are usually determined by the following factors:**

- How resistant the data is to becoming stale
- How sensitive the data is to user-specific customization

Creating and Storing Backups

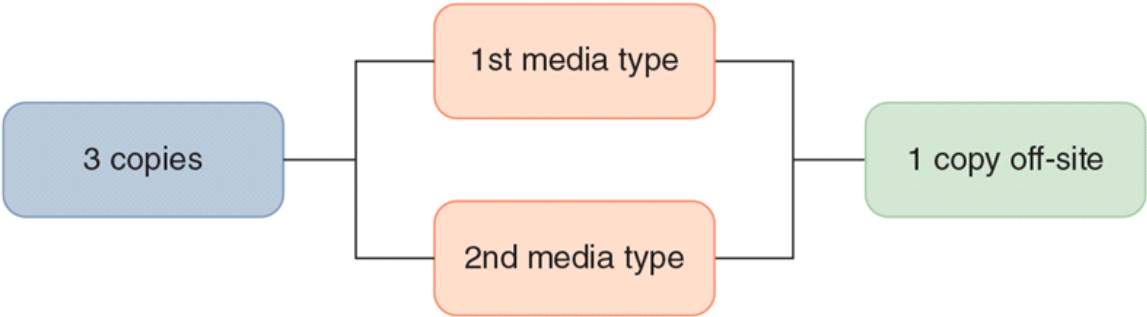


Figure 8-15 Three copies of data, stored on two different kinds of media, with one copy stored off-site

- **3 copies of data**
You should always have three copies of your data: one original and two backup copies.
- **2 different media types**
The data should be stored on two different types of storage media (for example: local disk and cloud storage).
- **1 copy off-site**
One copy must be stored in a different physical location to protect data from disasters like fire, theft, or system failure.

Why this is important

- Protects data against hardware failure
- Protects data against disasters
- Ensures data recovery in worst-case scenarios

Creating and Storing Backups

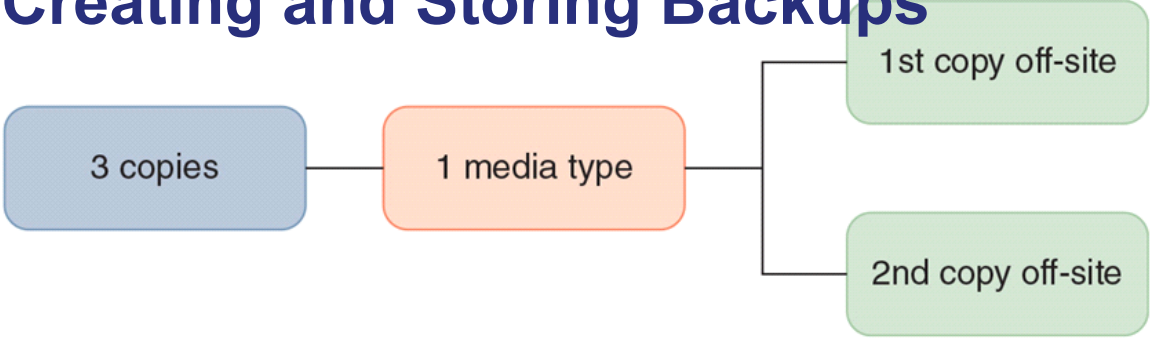


Figure 8-16 Three copies of data, stored on one kind of media (such as SSD), at two geographically isolated locations in the cloud

- **3 copies of data**
There are three copies of the same data.
- **1 media type**
All copies are stored on the same type of storage (for example, SSD storage in the cloud).
- **2 off-site copies**
Two copies are stored in different geographic locations in the cloud.

Why this is important

- Protects data from regional failures
- Uses geographic isolation instead of different media types
- Common in modern cloud backup systems

Protecting Capabilities

- Storing backups is a common use of cloud storage
- Cloud storage is often used to keep backup copies of data.
- Some common backup objects include the following:
 - Applications, file systems, configuration files, and databases
 - Consider the following protection capabilities:
 - **Network redundancy – HA (high availability)** means having reliable and continuous access to data
 - **Cloning** – Creating a new copy of existing data
 - **Storage replication** – Helps keep data durable by copying it to another location
 - **Storage mirroring** – Similar to replication, keeps an exact copy of data in another location

Backup Types (1 of 3)

- Different backup types copy data in different ways:
 - **Full backup**
Backs up all data every time a backup is created.
 - **Incremental backup**
Backs up only data that changed since the last backup of any type.
 - **Differential backup**
Backs up data that changed since the last full backup.
- **Snapshots**
 - A storage **snapshot** records the state of a storage system at a specific point in time.
 - It does not store the data itself, only metadata and pointers to the data blocks.